Preliminary Dense-RHS CoRL Compact Results

Autogenerated from local TD-MPC2-JAX run artifacts

Scope. This preliminary report summarizes the completed compact campaign "dense_rhs_corl_compact_v2" using local metric caches refreshed from the NCC run directories. It is intended as an internal CoRL-style evidence package, not as the final v3 matrix. The following environment is excluded from all results and plots: fish-swim.

Headline. Across 30 matched valid seed pairs, Adaptive RHS changes final score by **5.90%** ± **24.00%** relative to the fixed TD-MPC2 paper horizon, with an RHS win rate of **70.00%**. The summary excludes 0 missing, nonfinite, or unmatched pairs.

Experimental Setup

We compare the TD-MPC2 fixed paper horizon against Adaptive RHS under matched environment, regime, and seed cells. Environments are quadruped-run, cheetah-run, hopper-hop, finger-turn_hard, cartpole-swingup; regimes are clean and chaos; seeds are 1, 7, 15; training budget is 500000 environment steps. All rows come from `experiments/corl_compact_v2_ledger.csv`; each row records the SLURM job id, git commit, remote run directory, score, runtime, and checkpoint status.

Aggregate Result

Scope	n	RHS delta (%)	RHS wins	Runtime ratio
all valid pairs	30	5.90 ± 24.00	70.00%	1.13
chaos	15	11.43 ± 31.37	73.33%	1.13
clean	15	0.37 ± 11.99	66.67%	1.13

Full Matched Results

Environment	Regime	Seed	Fixed	RHS	Δ (%)	Fixed h	RHS h
cartpole-swingup	chaos	1	814.1	816.4	+0.28	1.76	2.04
cartpole-swingup	chaos	7	856.5	861.5	+0.59	1.78	2.15
cartpole-swingup	chaos	15	859.2	860.6	+0.16	1.81	2.18
cartpole-swingup	clean	1	882.1	864.9	-1.95	1.38	1.95
cartpole-swingup	clean	7	878.4	882.4	+0.45	1.37	1.62
cartpole-swingup	clean	15	864.5	879.8	+1.77	1.37	1.86
cheetah-run	chaos	1	354.5	501.1	+41.35	1.94	2.36
cheetah-run	chaos	7	515.4	595.9	+15.64	1.94	2.10
cheetah-run	chaos	15	528.2	540.9	+2.41	1.95	2.19
cheetah-run	clean	1	926.2	946.6	+2.20	1.51	1.62
cheetah-run	clean	7	944.0	942.7	-0.14	1.50	1.64
cheetah-run	clean	15	850.7	905.6	+6.45	1.50	1.63
finger-turn-hard	chaos	1	651.0	448.7	-31.08	1.95	2.05
finger-turn-hard	chaos	7	580.5	838.8	+44.50	1.94	2.21
finger-turn-hard	chaos	15	510.1	753.5	+47.71	1.97	2.06
finger-turn-hard	clean	1	772.7	918.8	+18.91	1.49	1.66
finger-turn-hard	clean	7	818.2	917.0	+12.07	1.50	1.60
finger-turn-hard	clean	15	884.6	928.9	+5.01	1.50	1.60
hopper-hop	chaos	1	166.8	278.1	+66.71	2.15	2.46
hopper-hop	chaos	7	292.4	159.1	-45.59	2.14	2.58
hopper-hop	chaos	15	158.2	139.0	-12.17	2.17	2.55
hopper-hop	clean	1	542.8	570.1	+5.04	1.72	1.83
hopper-hop	clean	7	514.9	469.7	-8.78	1.71	1.86
hopper-hop	clean	15	559.3	357.9	-36.00	1.73	1.87
quadruped-run	chaos	1	660.8	732.9	+10.90	2.43	2.64

Environment	Regime	Seed	Fixed	RHS	Δ (%)	Fixed h	RHS h
quadruped-run	chaos	7	807.8	722.1	-10.60	2.41	2.56
quadruped-run	chaos	15	576.1	810.0	+40.59	2.42	2.56
quadruped-run	clean	1	881.8	913.6	+3.60	1.93	2.13
quadruped-run	clean	7	879.3	839.9	-4.48	1.96	2.10
quadruped-run	clean	15	911.0	923.8	+1.41	1.99	2.25

Learning Curves

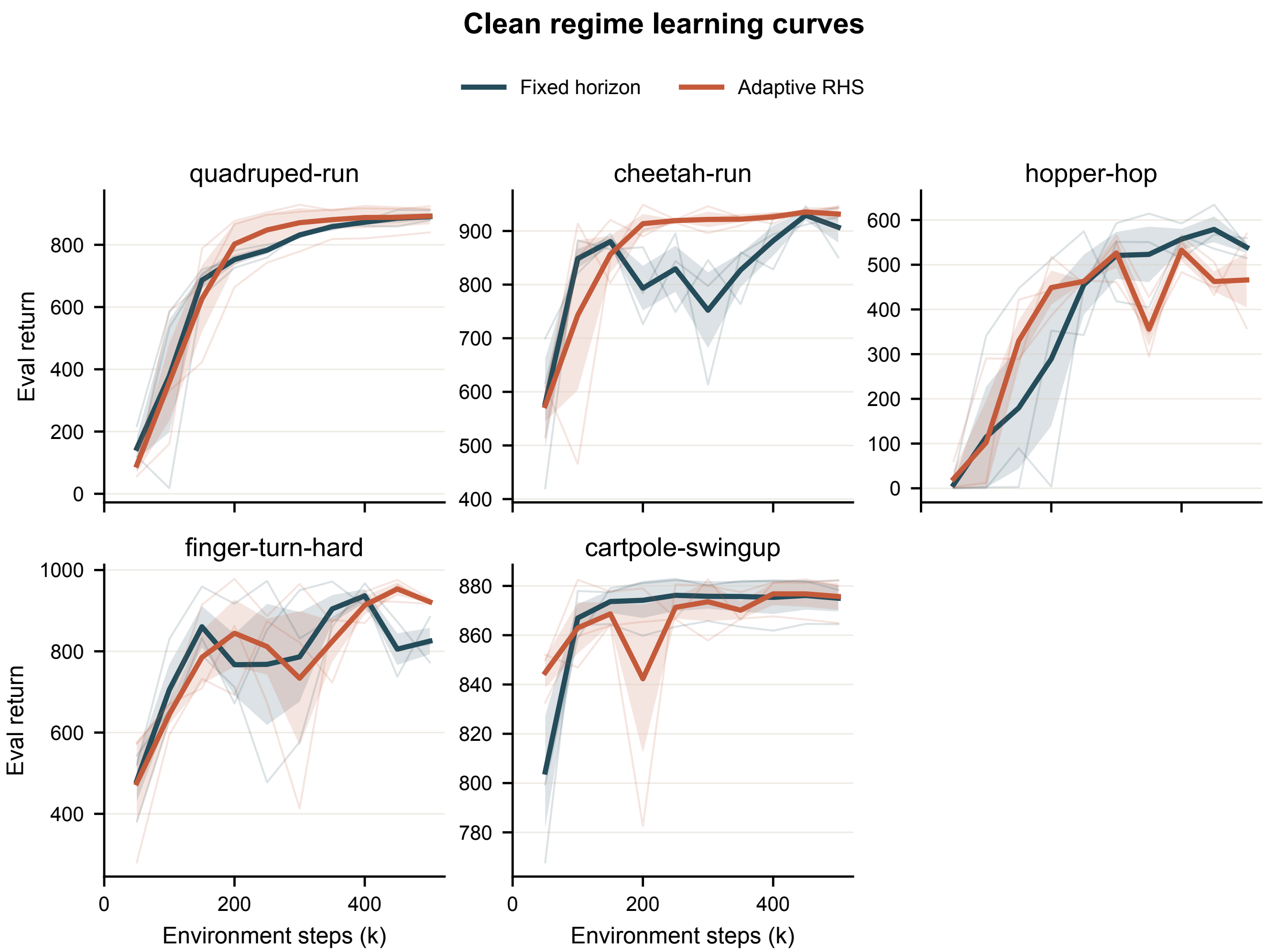

Figure 1: Clean-regime evaluation return. Thin traces are individual seeds; bold traces show the seed mean with standard error.

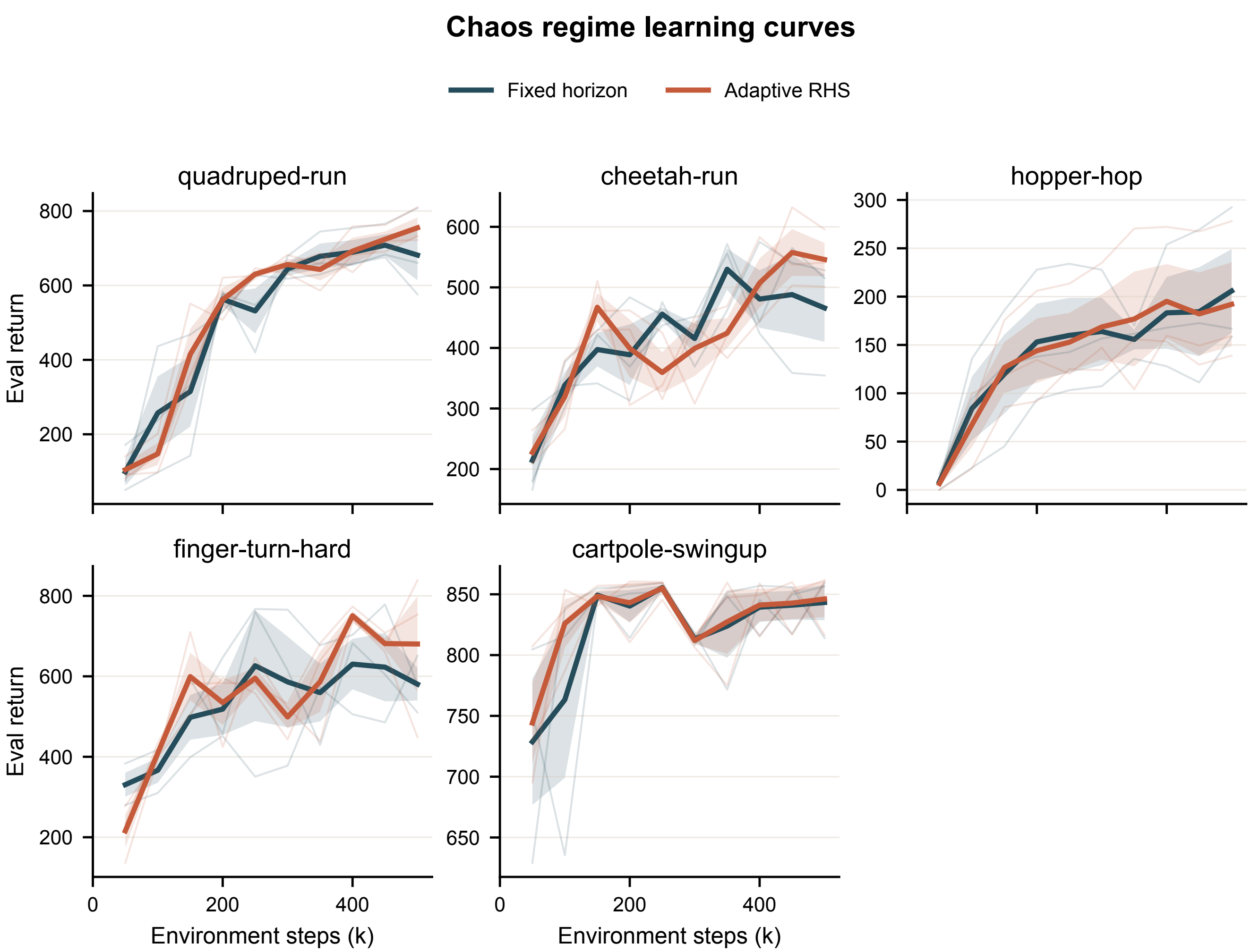

Figure 2: Chaos-regime evaluation return under matched randomized/noisy evaluation.

Matched Percent Delta

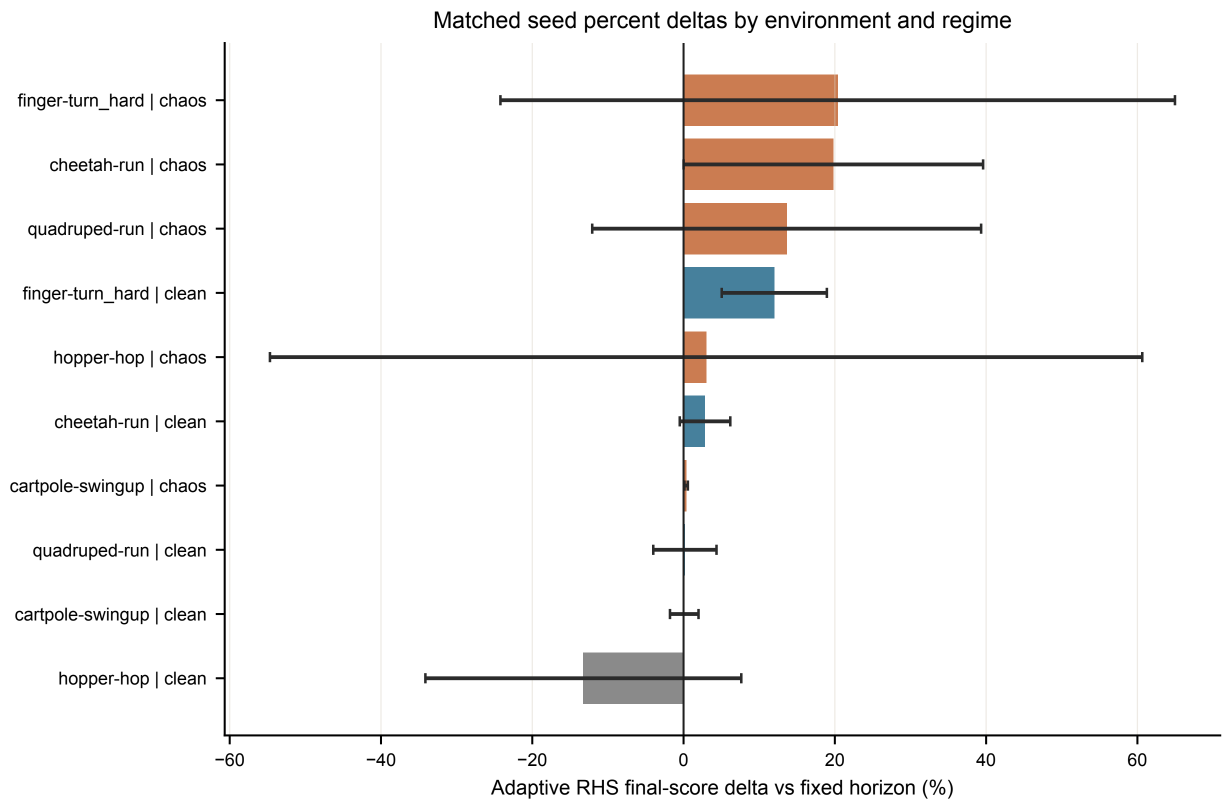

Figure 3: Per-environment and per-regime Adaptive RHS percent delta against the matched fixed-horizon baseline. Error bars are across seeds.

Environment	Regime	n	RHS delta (%)	RHS wins	RHS h
cartpole-swingup	chaos	3	0.34 ± 0.22	100.0%	2.12
cartpole-swingup	clean	3	0.09 ± 1.89	66.67%	1.81
cheetah-run	chaos	3	19.80 ± 19.80	100.0%	2.22
cheetah-run	clean	3	2.84 ± 3.34	66.67%	1.63
finger-turn_hard	chaos	3	20.38 ± 44.59	66.67%	2.11
finger-turn_hard	clean	3	11.99 ± 6.95	100.0%	1.62
hopper-hop	chaos	3	2.98 ± 57.66	33.33%	2.53
hopper-hop	clean	3	-13.25 ± 20.88	33.33%	1.85
quadruped-run	chaos	3	13.63 ± 25.71	66.67%	2.59
quadruped-run	clean	3	0.18 ± 4.18	66.67%	2.16

Efficiency and Pareto View

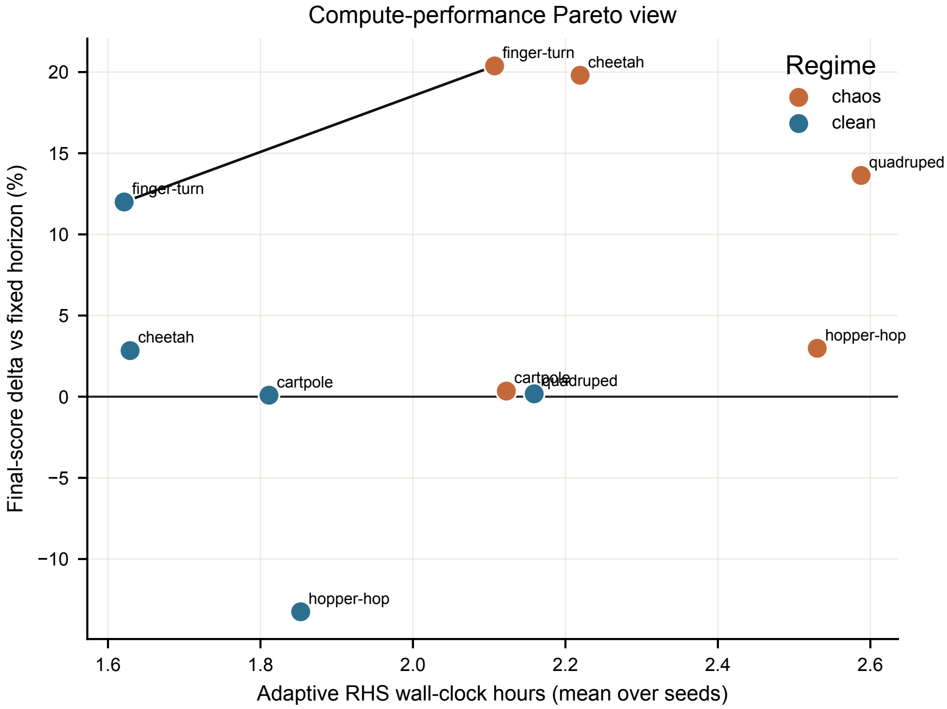

Figure 4: Compute-performance Pareto view. The x-axis is mean Adaptive RHS wall-clock hours and the y-axis is normalized final-score delta versus the matched fixed-horizon baseline; the black line highlights nondominated points.

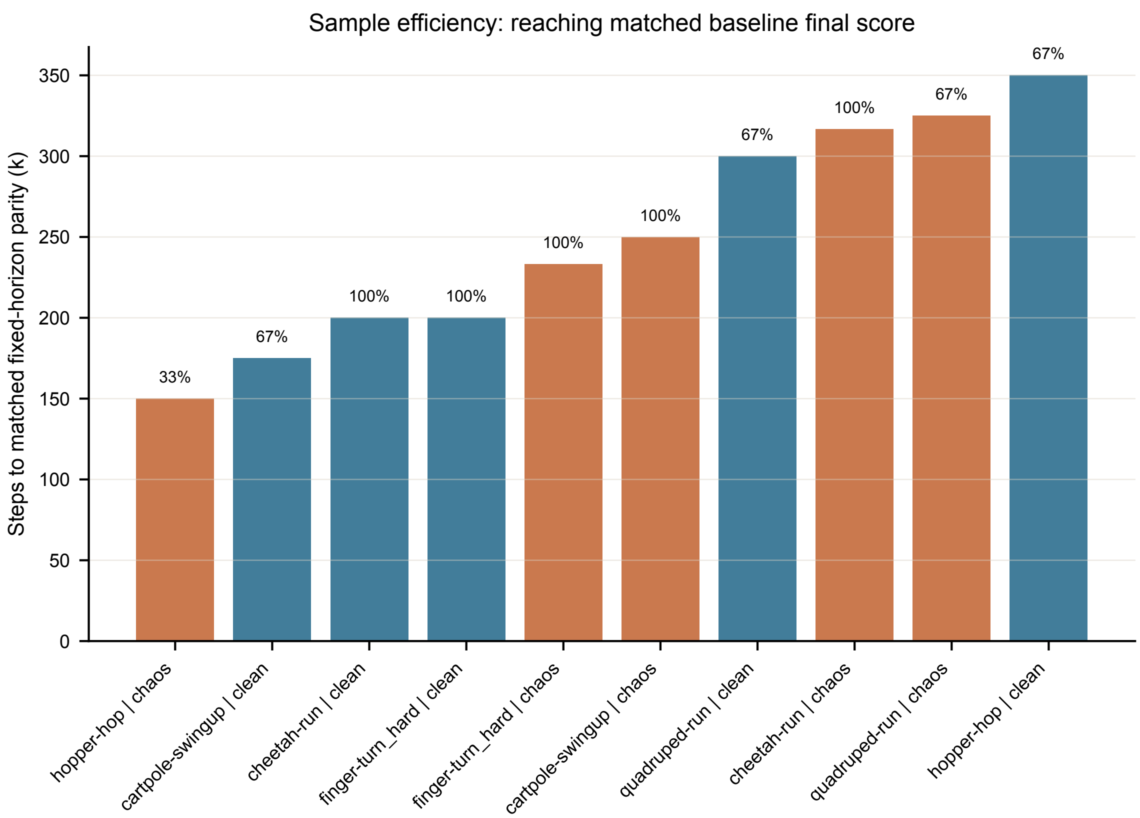

Figure 5: Sample efficiency, measured as the first Adaptive RHS evaluation point that reaches the matched fixed-horizon final score. Labels above bars indicate the fraction of seeds reaching parity.

RHS Diagnostics

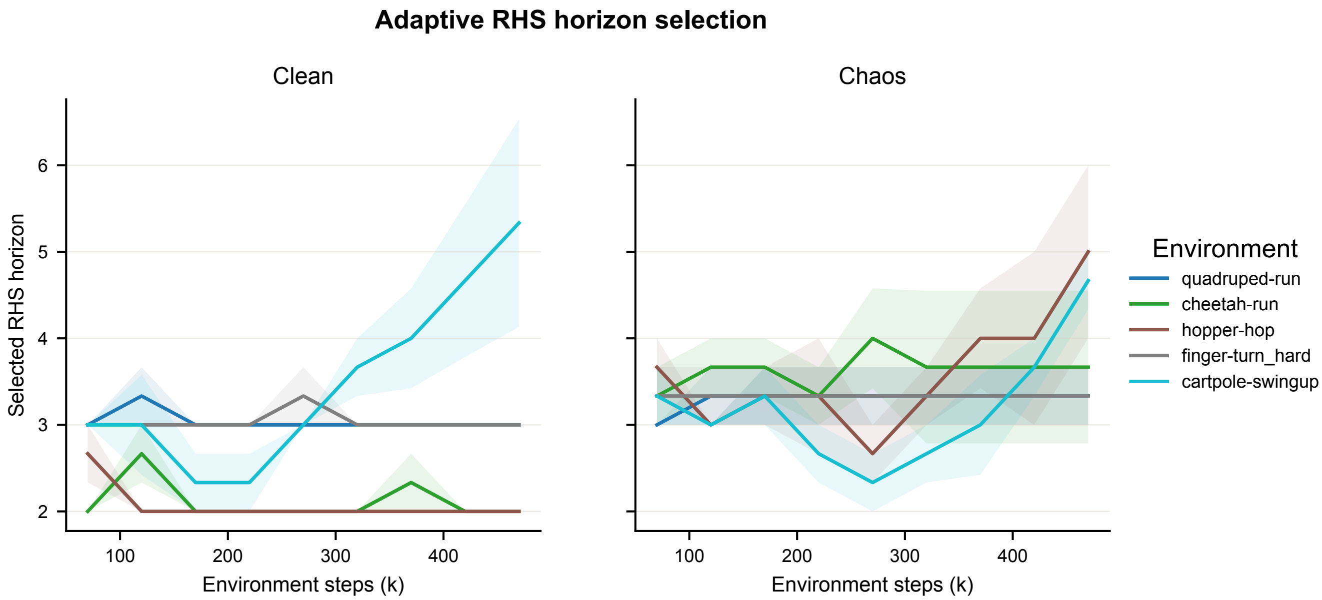

Figure 6: Adaptive RHS selected horizon over training, averaged across seeds.

Limitations

This is a preliminary report for the completed compact run after applying the report-level environment filter (60 retained terminal rows). Fish-swim is intentionally omitted because it was later removed from the v3 final matrix after isolated MJX chaos-push diagnostics exposed nonfinite state/reward failures. The final v3 package should reuse the same report template with the updated environment set and six seeds.

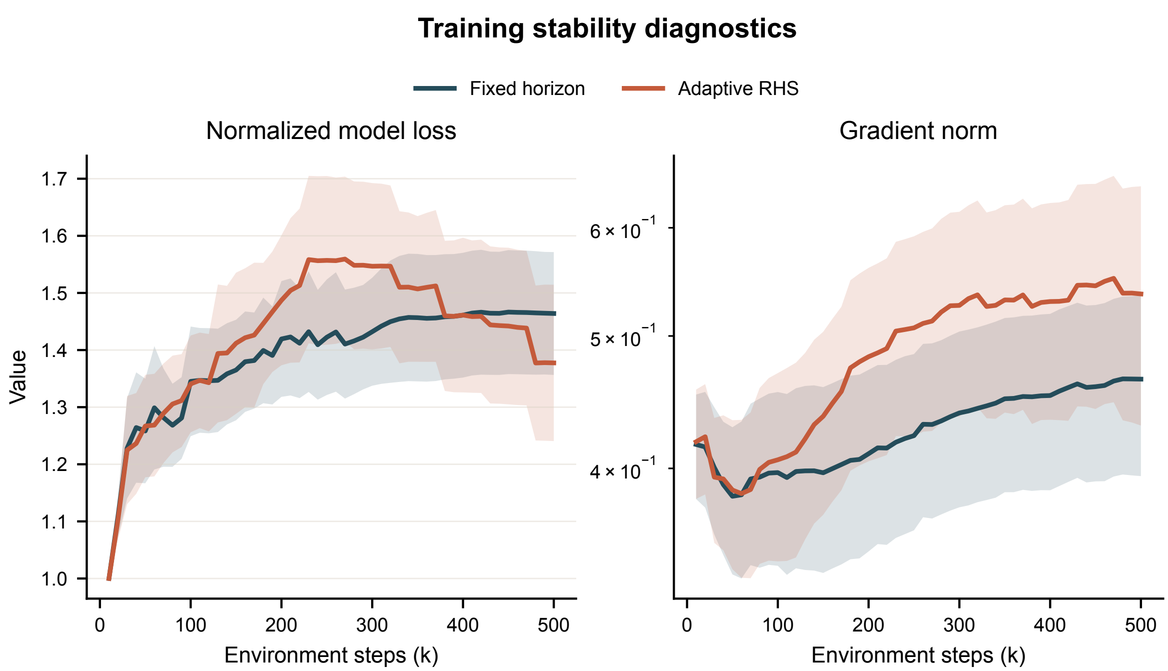

Figure 7: Training stability diagnostics pooled across runs. Total loss is normalized per run by its first finite value; gradient norms use a log axis.